Peer review and response

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Teach the evidence-to-conclusion chain for Peer review and response

Address the concern, preserve original records and distinguish proposed analysis from completed evidence.

Keep one habit: Model the main case → reveal the picture gradually → probe limits → assess independent reasoning → plan a changed-case return.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of peer review and response

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Writing results and limitations. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of writing results and limitations

Reference example, not the only acceptable wording: Teacher explanation: Results and limitations should state observed estimates before interpreting them. Separate measured comparisons from causal or deployment claims.

Give counts, denominators, effect sizes, valid uncertainty and failed conditions. State missing **evidence** and preserve the original primary outcome.

Without opening the old lesson, explain one key idea from Figures that preserve **evidence**. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of figures that preserve **evidence**

Reference example, not the only acceptable wording: Teacher explanation: Figures should preserve denominators, scales and uncertainty rather than exaggerate the preferred comparison.

A plot is a representation of **evidence**, not extra **evidence**. Label axes, units, cases and scope. For simple comparative bars use a zero baseline or explicitly justify another plot.

Show condition breakdowns and unassessed counts where relevant.

Without opening the old lesson, explain one key idea from Inspecting failures and anomalies. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of inspecting failures and anomalies

Reference example, not the only acceptable wording: Teacher explanation: Failure inspection generates explanations to test. An unusual result is not automatically a bad data point.

Compare raw **output**s, **input** validity and procedure traces. Correct demonstrable measurement errors with provenance; retain valid unusual **observation**s. Label post-result investigations exploratory.

Pictured end goal

Explain peer review and response using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Writing results and limitations

Figures that preserve evidence

Inspecting failures and anomalies

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. Peer review challenges **evidence**, method and clarity. Responses should explain changes with artifacts rather than merely defend the original wording.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of peer review and response

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model the whole case and check each inference

> Keep the independent answer private until after the first attempt.

Definitions and complete contract

Peer review challenges **evidence**, method and clarity. Responses should explain changes with artifacts rather than merely defend the original wording.

For each comment record concern, response, change location and new **evidence** if needed. If no new **evidence** exists, narrow the claim instead of promising an unperformed test.

Review item	Complete supplied exercise contract
Main concern	Fitted-line baseline comparison is missing
Main response	Neural network is advanced, without comparison results
Evidence status	No new baseline counts, design or raw outcomes supplied
Main repair options	Fair comparison under an explicit plan, labelled new analysis; or narrower claim
Separate picture	Four views of P1 plus four views of P2
Underlying source identities	Eight photos, two plants; not eight independent plants
Review artifact	Concern, response, change location, evidence and status

Review item	Complete supplied exercise contract
Limits	No actual reviewer communication, new experiment or source-independence proof

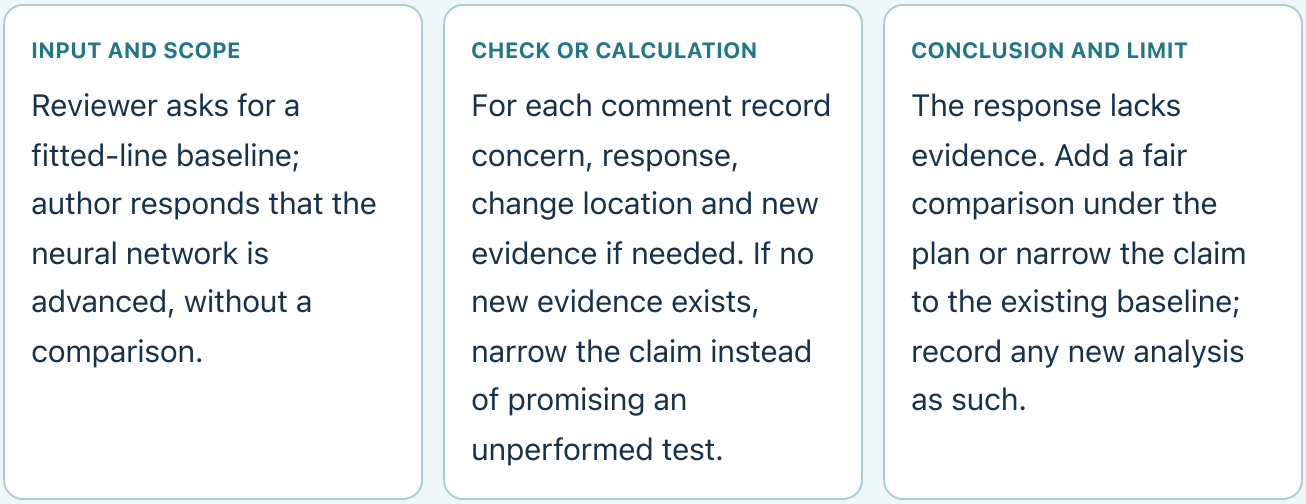
Main evidence ledger and checked key

Main response step	Evidence-grounded alternative
Identify concern	Does performance differ from a fitted-line baseline under aligned conditions?
Reject unsupported defense	Advanced architecture is not comparative evidence
If no comparison exists	Narrow the claim to the actual recorded comparison
If new comparison is proposed	Specify aligned cases/metric/eligibility and label new analysis
Change location	Identify the actual manuscript section/artifact when revised; do not invent it
Status	Proposed, completed or blocked, with corresponding evidence
Preserve history	Original claim, response and later analysis remain distinguishable

Main advanced-network wording does not answer the baseline request. Require a fair documented comparison labelled new work, or a narrower claim using actual evidence; do not fabricate results. Picture has eight photos but only two plant IDs: four views each. Revisit grouping, role allocation and uncertainty assumptions that treated views as independent plants. Word replacement alone fails.

No revised outcome rate or actual split leakage is supplied. Independent response should report actual failure counts with denominators, locations and evidence, preserve relevant limitations and disclose any missing records. Deleting limitations does not answer the concern. No actual communication, completed experiment or independence guarantee occurs.

Peer review and response — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: The response lacks **evidence**. Add a fair comparison under the plan or narrow the claim to the existing baseline; record any new analysis as such.

Qualification: Agreement with a reviewer is not required, but a disagreement needs explicit reasoning and **evidence**.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Reviewer asks for a fitted-line baseline; author responds that the neural network is advanced, without a comparison.

The response lacks **evidence**. Add a fair comparison under the plan or narrow the claim to the existing baseline; record any new analysis as such.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Boundary and misconception probes

Review-response audit	Checked interpretation
Reviewer asks for baseline	Answer with a comparison or a narrower claim
Author says advanced	Architecture praise is not evidence
New analysis planned	Keep status proposed until artifacts/results exist
Eight photos from two plants	Do not count eight independent plants
Correct the word only	Does not repair downstream unit assumptions
Two plant IDs	Not proof of independent sampling
Missing failure counts	Provide actual counts/denominators or disclose absence
Delete limitation paragraph	Hides concern rather than resolving it

Separate pictured facts and full reasoning

Respond to a reviewer with **evidence**. A reviewer asks why four photos of one plant were treated as four independent plants.

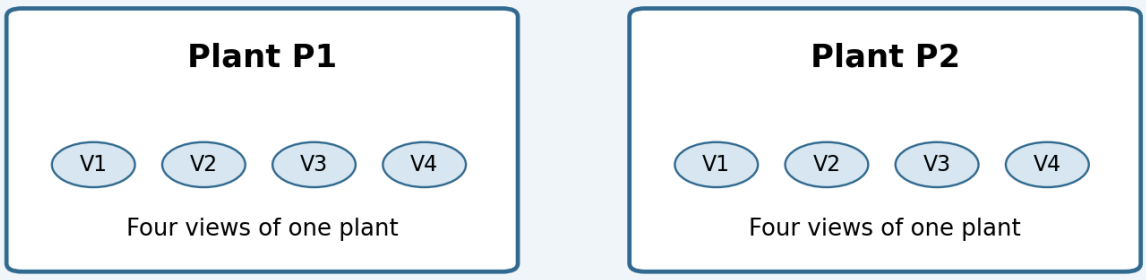
The original table contains four views of P1 and four views of P2: eight photos from two plants.

The separate picture has four views of plant P1 and four of P2, so eight photos represent two distinct plant IDs. A claim of eight independent plants is unsupported. Changing the word photos to plants does not change the source structure. Revisit the sampling/inference unit, group allocation across data roles and any uncertainty calculation that assumed eight independent plant observations.

The picture supplies no correctness labels or split assignment, so neither a revised accuracy nor actual leakage can be inferred. Correct grouping does not create more plants, and two different IDs do not prove independent sampling.

Source grouping	Photo views	Plant identities
P1	V1, V2, V3, V4	One
P2	V1, V2, V3, V4	One
Total	Eight photos	Two distinct plant IDs
Unsupported analysis premise	Eight independent plants	Source structure contradicts this count
Required reconsideration	Grouping, split roles and uncertainty assumptions	Not just a word replacement
Outcome/split evidence	Not supplied	No invented revised score or known leakage

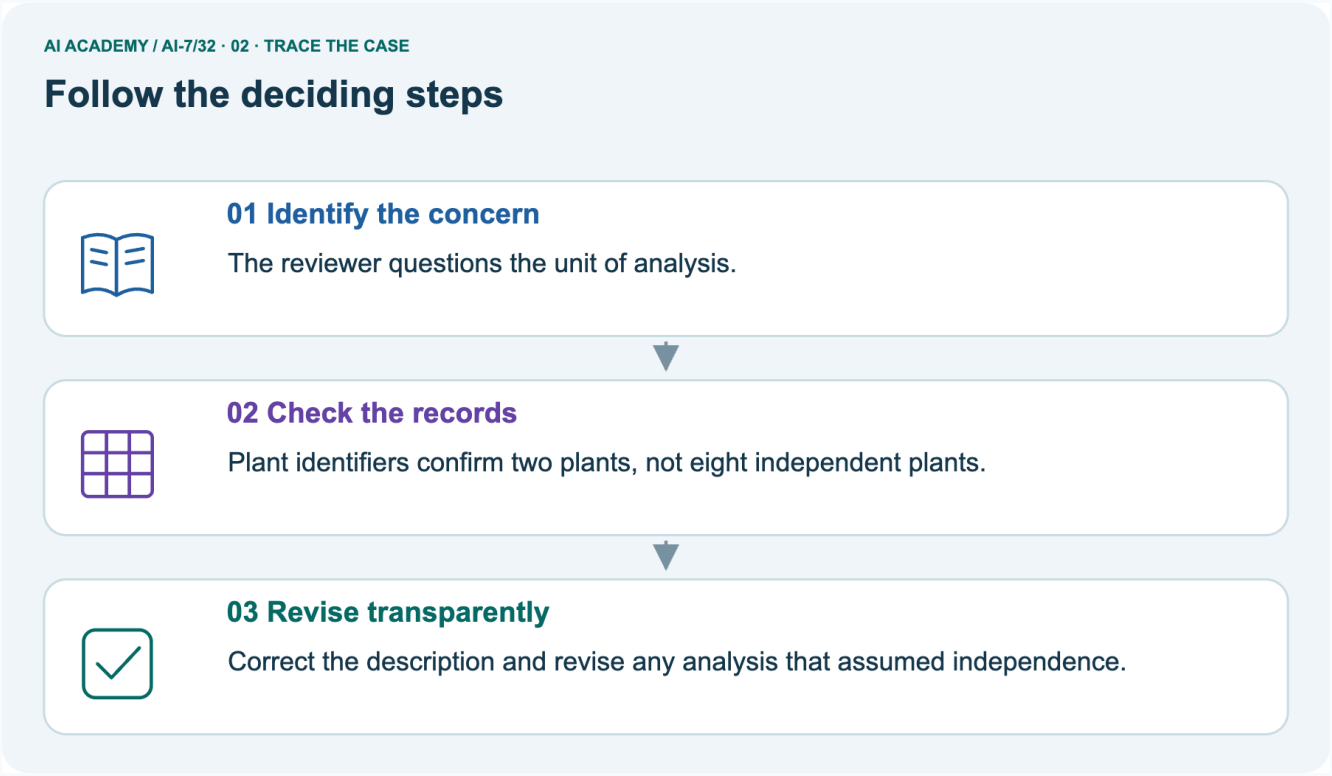
Repair the unit assumption, not just the wording



Eight photos → two distinct plant identities

Revisit grouping, split roles and uncertainty assumptions; no outcome counts are supplied. Two IDs do not prove independent sampling. Changing labels alone cannot fix the analysis.

Picture trace and changed-condition key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Identify the concern	The reviewer questions the unit of analysis.
2. Check the records	Plant identifiers confirm two plants, not eight independent plants.
3. Revise transparently	Correct the description and revise any analysis that assumed independence.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Peer review should connect a criticism to the actual records and a concrete correction or reasoned response. Polite language alone cannot repair a methodological error.

Two plants provide limited coverage; grouping correctly does not create more independent **evidence**.

Change: The team only changes "photos" to "plants" without revisiting analysis.

Prediction: The substantive independence issue remains unresolved.

Reason: Renaming records does not change their shared source objects.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Review comment	A concern about evidence, method or presentation
Response record	A mapping from comment to change or justified disagreement
Claim narrowing	Reducing a statement to the evidence actually available

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

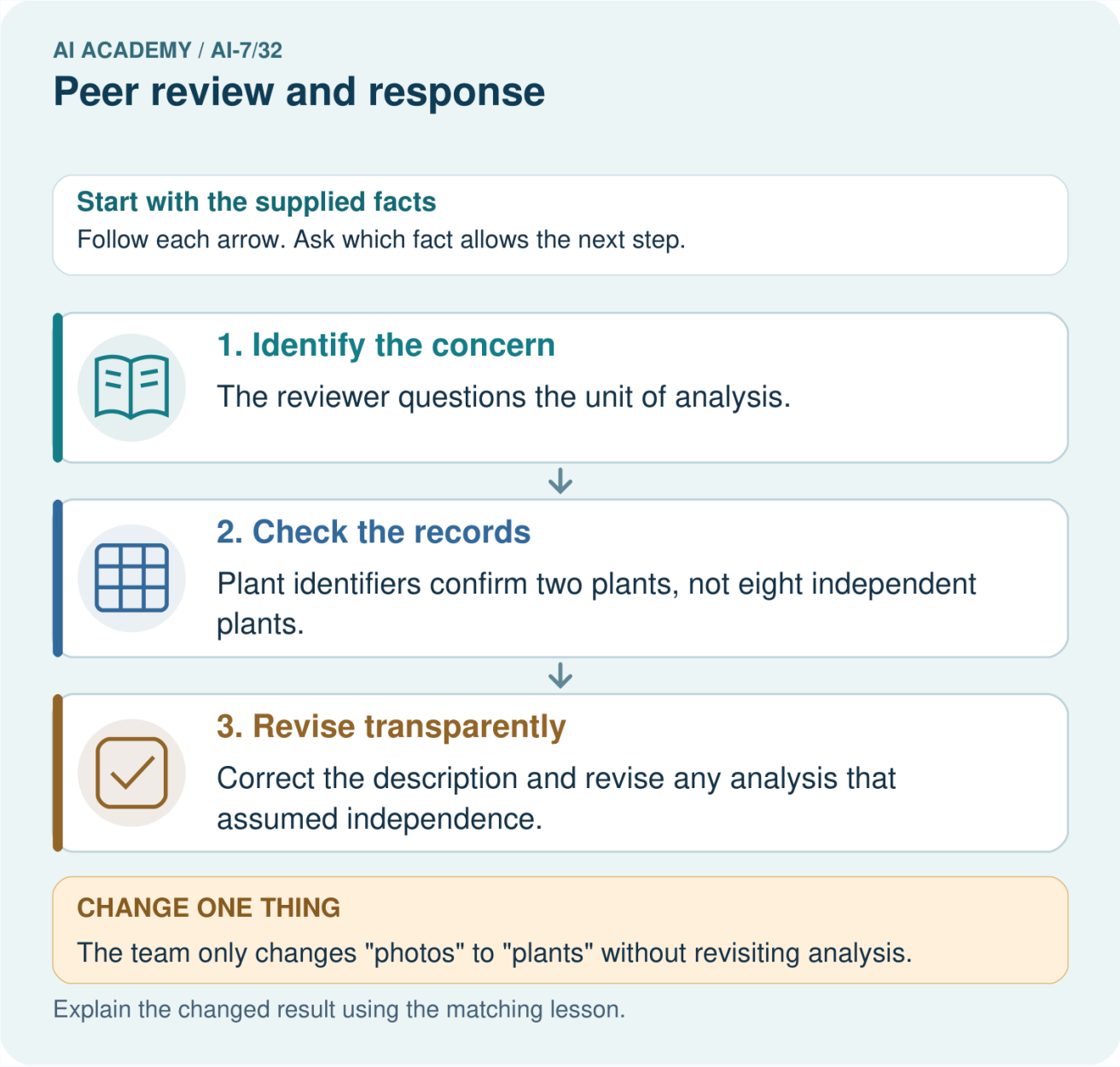
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Peer review and response

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Reviewer asks for a fitted-line baseline; author responds that the neural network is advanced, without a comparison. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: The response lacks **evidence**. Add a fair comparison under the plan or narrow the claim to the existing baseline; record any new analysis as such.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Peer review and response. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Agreement with a reviewer is not required, but a disagreement needs explicit reasoning and **evidence**.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Count photos and underlying plants in the reviewer's concern. What analysis assumption needs revision, and why does changing only the word photos to plants fail to repair it?

The reviewer questions the unit of analysis. Plant identifiers confirm two plants, not eight independent plants. Correct the description and revise any analysis that assumed independence.

Changed case: The substantive independence issue remains unresolved. Renaming records does not change their shared source objects.

Changed-case reasoning: The substantive independence issue remains unresolved. Renaming records does not change their shared source objects.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main advanced-network wording does not answer the baseline request. Require a fair documented comparison labelled new work, or a narrower claim using actual evidence; do not fabricate results. Picture has eight photos but only two plant IDs: four views each. Revisit grouping, role allocation and uncertainty assumptions that treated views as independent plants. Word replacement alone fails.

No revised outcome rate or actual split leakage is supplied. Independent response should report actual failure counts with denominators, locations and evidence, preserve relevant limitations and disclose any missing records. Deleting limitations does not answer the concern. No actual communication, completed experiment or independence guarantee occurs.

Diagnose the current misconceptions

The worked result establishes every future case: Agreement with a reviewer is not required, but a disagreement needs explicit reasoning and **evidence**.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows	Next teaching action
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Reviewer notes missing failure counts; author deletes the limitation paragraph. What is an adequate response?

Teacher explanation: Supply counts and denominators with raw **evidence**, retain the limitation and revise the claim. Deleting criticism does not resolve the **evidence** gap.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Peer review challenges **evidence**, method and clarity. Responses should explain changes with artifacts rather than merely defend the original wording.

For each comment record concern, response, change location and new **evidence** if needed. If no new **evidence** exists, narrow the claim instead of promising an unperformed test.

W2 Trace the worked case

Reviewer asks for a fitted-line baseline; author responds that the neural network is advanced, without a comparison. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: The response lacks **evidence**. Add a fair comparison under the plan or narrow the claim to the existing baseline; record any new analysis as such.

W3 Audit the limit

A report omits this qualification: Agreement with a reviewer is not required, but a disagreement needs explicit reasoning and **evidence**.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Agreement with a reviewer is not required, but a disagreement needs explicit reasoning and **evidence**.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Agreement with a reviewer is not required, but a disagreement needs explicit reasoning and **evidence**.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: The response lacks **evidence**. Add a fair comparison under the plan or narrow the claim to the existing baseline; record any new analysis as such.

Qualification: Agreement with a reviewer is not required, but a disagreement needs explicit reasoning and **evidence**.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Reviewer notes missing failure counts; author deletes the limitation paragraph. What is an adequate response?

Teacher explanation: Supply counts and denominators with raw **evidence**, retain the limitation and revise the claim. Deleting criticism does not resolve the **evidence** gap.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Supply counts and denominators with raw **evidence**, retain the limitation and revise the claim. Deleting criticism does not resolve the **evidence** gap. Require intermediate reasoning and the stated scope.

Fresh retry: Reviewer asks whether timing includes retrieval; record does not specify. Expected reasoning: State the timing boundary if recoverable from records; otherwise mark speed comparison unresolved and narrow the claim.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled peer review and response **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of peer review and response.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Independent reproduction check. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/33, ai-7/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of peer review and response

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Reviewer asks for a fitted-line baseline; author responds that the neural network is advanced, without a comparison.

Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: The response lacks **evidence**. Add a fair comparison under the plan or narrow the claim to the existing baseline; record any new analysis as such.

Check every learner before discussion: Use the worked case for Peer review and response. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Agreement with a reviewer is not required, but a disagreement needs explicit reasoning and **evidence**.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The team only changes "photos" to "plants" without revisiting analysis. Require a **prediction** and a reason before comparing. Expected result: The substantive independence issue remains unresolved.

Reason: Renaming records does not change their shared source objects.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

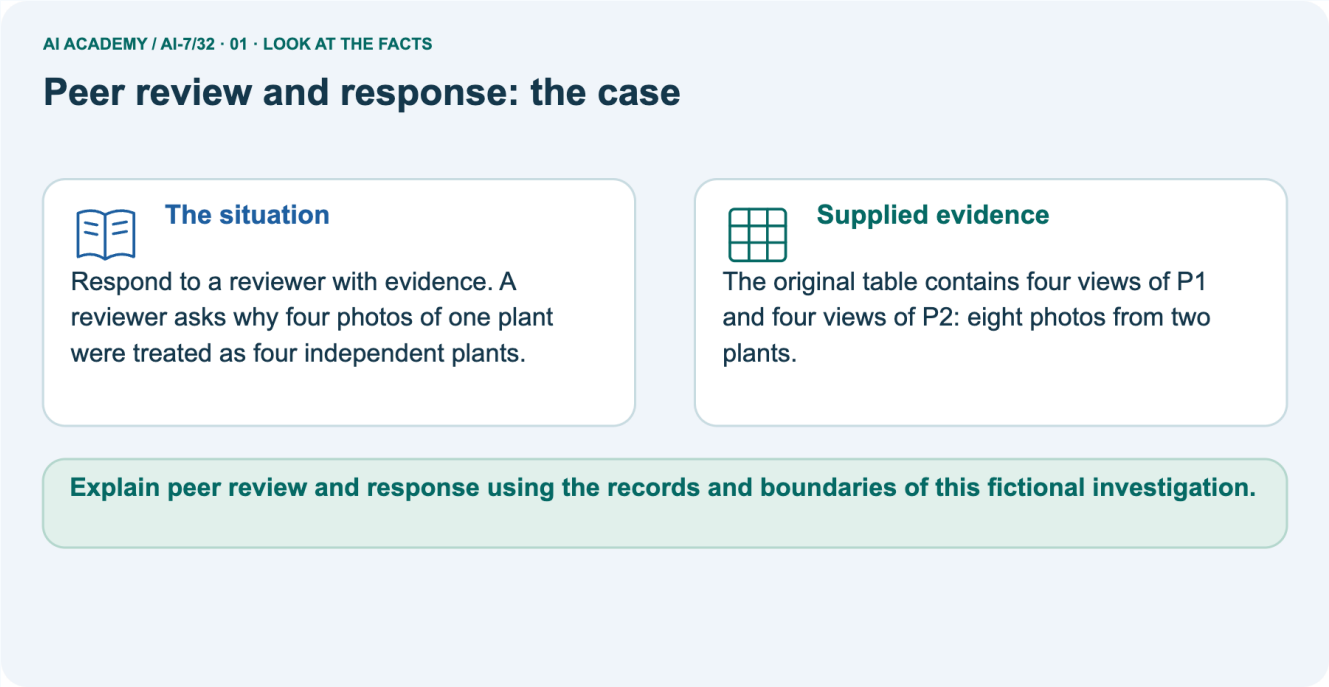
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Peer review and response: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

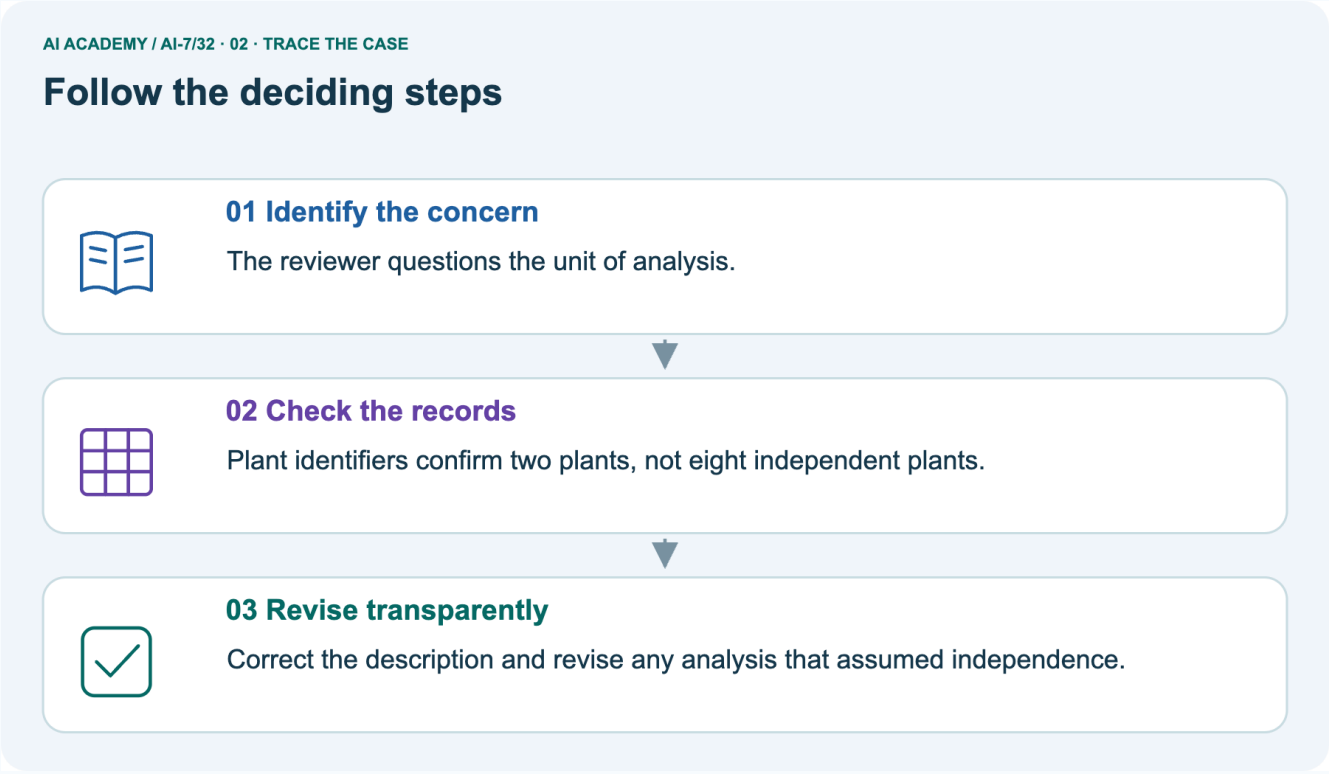
Respond to a reviewer with **evidence**. A reviewer asks why four photos of one plant were treated as four independent plants.

Supplied facts: The original table contains four views of P1 and four views of P2: eight photos from two plants.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



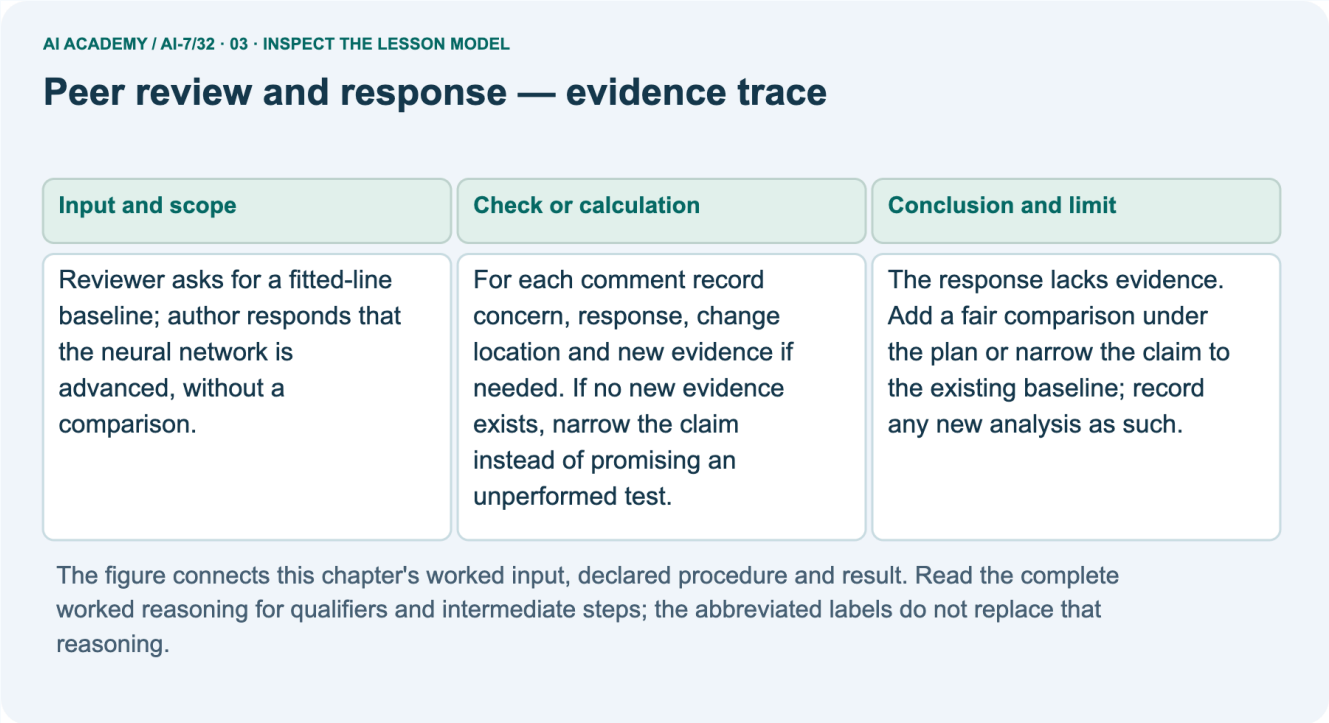
Follow the labelled picture with the matching explanation. Open the vector diagram

Identify the concern: The reviewer questions the unit of analysis. → Check the records: Plant identifiers confirm two plants, not eight independent plants. → Revise transparently: Correct the description and revise any analysis that assumed independence.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Peer review and response — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Peer review and response — **evidence** trace.

Reviewer asks for a fitted-line baseline; author responds that the neural network is advanced, without a comparison. | For each comment record concern, response, change location and new **evidence** if needed.

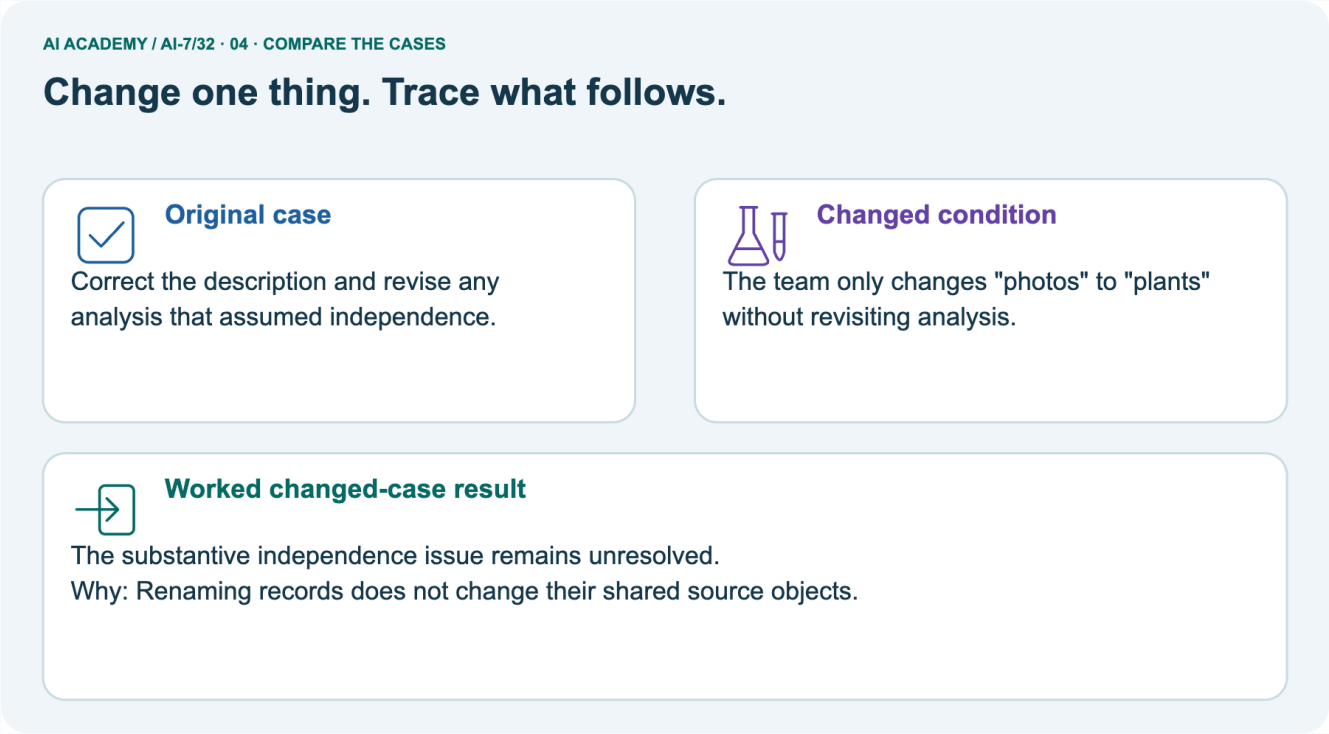
If no new **evidence** exists, narrow the claim instead of promising an unperformed test. | The response lacks **evidence**.

Add a fair comparison under the plan or narrow the claim to the existing baseline; record any new analysis as such.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Correct the description and revise any analysis that assumed independence. Changed condition: The team only changes "photos" to "plants" without revisiting analysis.

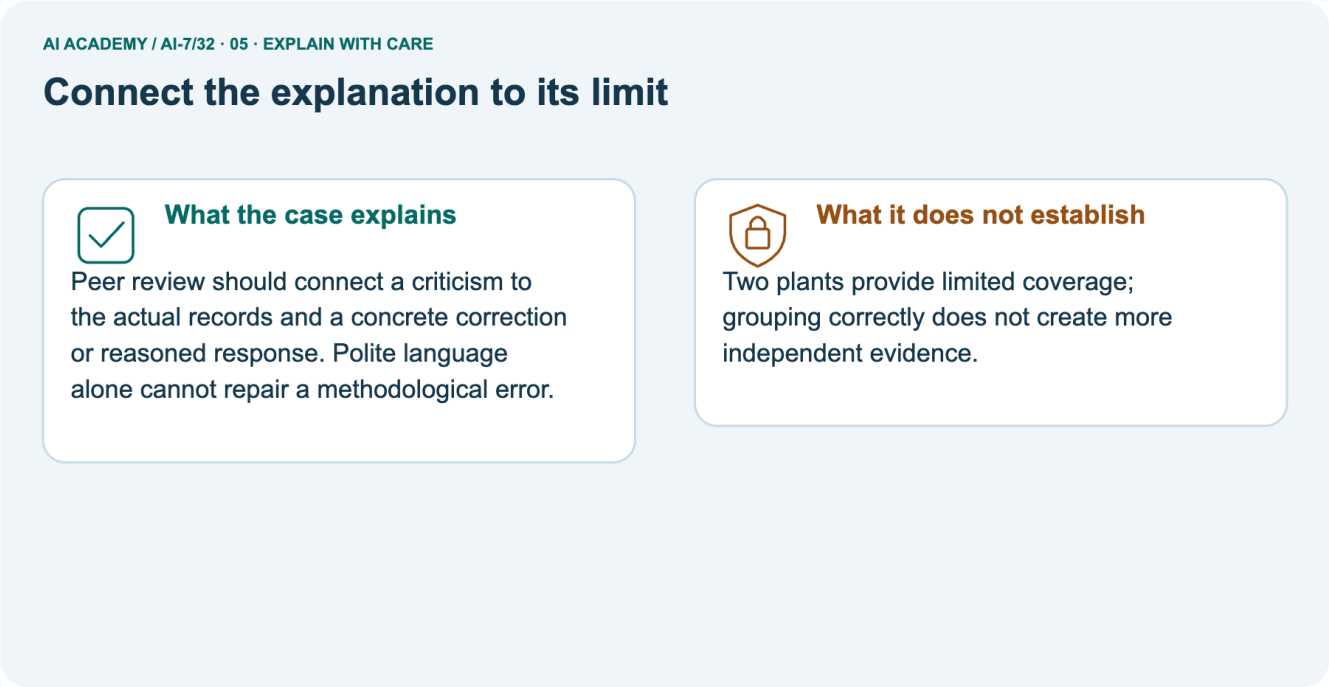
Expected result: The substantive independence issue remains unresolved. Reason: Renaming records does not change their shared source objects.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Renaming records does not change their shared source objects.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Peer review should connect a criticism to the actual records and a concrete correction or reasoned response. Polite language alone cannot repair a methodological error.

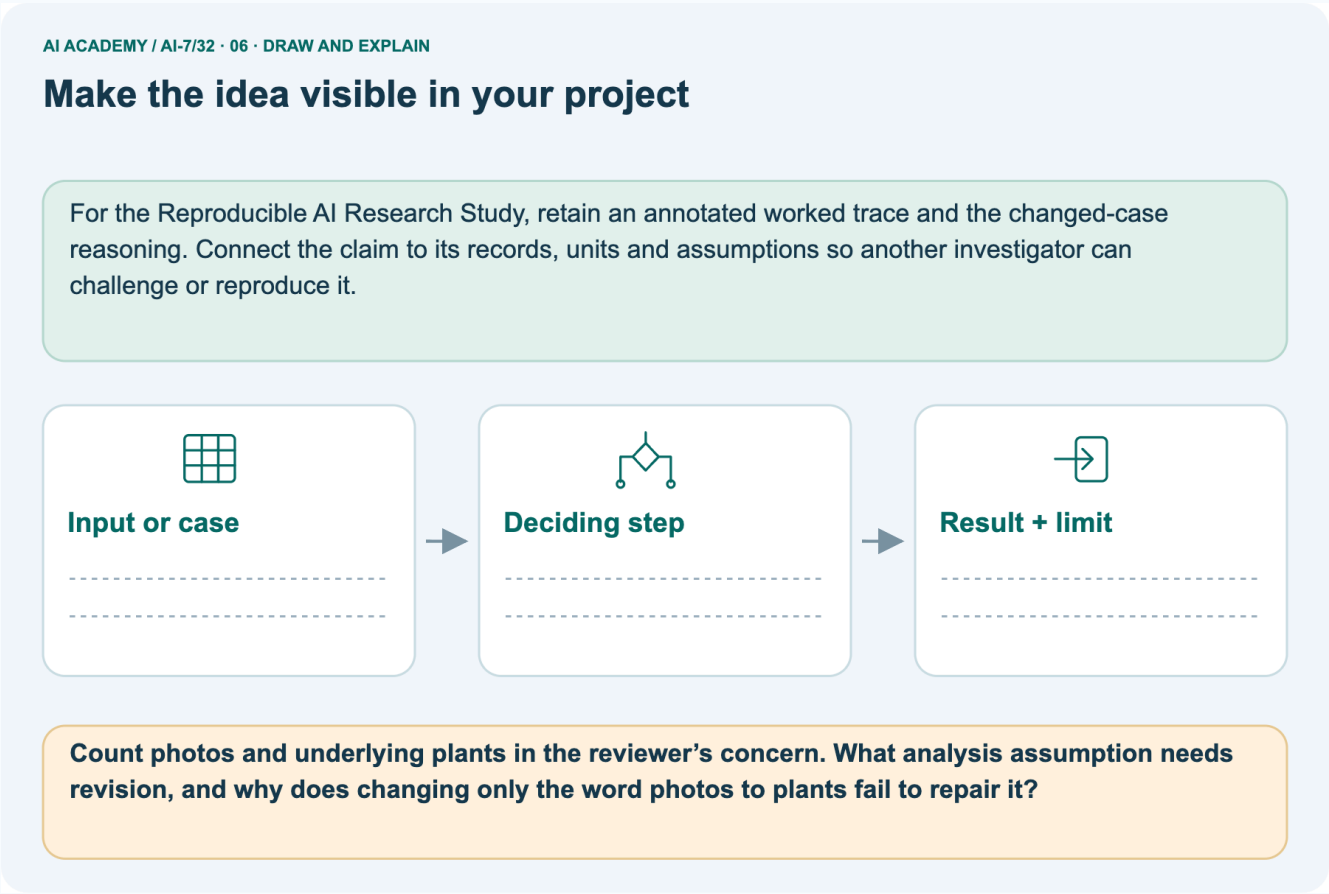
Limit: Two plants provide limited coverage; grouping correctly does not create more independent **evidence**.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Two plants provide limited coverage; grouping correctly does not create more independent **evidence**.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Count photos and underlying plants in the reviewer's concern.

What analysis assumption needs revision, and why does changing only the word photos to plants fail to repair it?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The reviewer questions the unit of analysis. Plant identifiers confirm two plants, not eight independent plants. Correct the description and revise any analysis that assumed independence.

Changed case: The substantive independence issue remains unresolved. Renaming records does not change their shared source objects.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Peer review and response: the case

02. Follow the deciding steps

03. Peer review and response — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.